

A Minimum Discrepancy Method for Weibull Distribution Parameter Estimation

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The most applied principles for parameter estimation are maximum likelihood and least square error. This paper presents a new principle with regard to the parameter estimation of the three-parameter Weibull distribution. By transforming the cumulative distribution function, constructed is a mapping from the value of the random variable and its corresponding cumulative distribution probability to the scale parameter. The scale parameter estimated by such a mapping is the random variable value and the corresponding cumulative distribution probability dependent when the shape parameter and/or location parameter applied in the mapping is subject to error. Given a set of random variable values or a set of sample values, a larger error in the shape/location parameter brings about larger differences between the scale parameter values obtained with the individual random variable values or sample values, respectively. Based on such a causal relationship between the discrepancy and the shape/location parameter value applied in the mapping relation, a new parameter estimation method is proposed. For the Weibull distribution parameter estimation according to a set of sample values, the right shape parameter and location parameter are those minimizing the discrepancy between the scale parameter values obtained with the individual sample values, respectively. Case studies demonstrate that the proposed method outperforms the maximum likelihood method and the Weibull plot-based least squares method.

Keywords: Three-parameter Weibull distribution; parameter estimation; pseudo estimator; parameter estimation principle; parameter estimation criterion.

1. Introduction

The Weibull distribution has been widely used to describe product lifetimes because of its mathematical tractability and adaptability. The majority of random variables

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involved in engineering problems can be well described with Weibull distribution, except for some special cases.^{1,2} Nevertheless, the mostly applied is the two-parameter Weibull distribution.³⁻⁶ The three-parameter Weibull distribution has more advantages, especially for describing long life and high reliability product.⁷⁻⁹ However, more parameters in a probability distribution function mean more complications for parameter estimation. Unless the sample size is large enough, the error of the conventional parameter estimation methods for three-parameter Weibull distributions is quite large and unstable.¹⁰

Parameter estimation for the three-parameter Weibull distribution has long been a concern for both methodological research and practical application.¹¹⁻¹³ Lots of efforts have been made to propose or revise parameter estimation methods.¹⁴⁻²¹ To overcome the shortcomings of the traditional maximum likelihood method in terms of difficult equation-resolving and non-regularity, a variety of alternative approaches were developed.²²⁻²⁵

Nagatsuka⁹ proposed an efficient method for the estimation of parameters and quantiles of the three-parameter Weibull distribution by using statistics invariant to the unknown location parameter. To estimate the location parameter of the three-parameter Weibull distribution, Sugiyama²⁶ proposed a minimum-variance linear unbiased estimator based on the order statistics. Ouedraogo *et al.* proposed a differential evolution optimization method.²⁷ Jukic *et al.* proposed a non-linear weighted errors-in-variables fitting approach.²⁸ Acitas *et al.* proposed a particle swarm optimization method with a new adaptive search space obtained by the modified maximum likelihood estimators.²⁹ Yonar *et al.* proposed a heuristic method by means of the dragonfly algorithm for obtaining the maximum likelihood estimates of the three-parameter Weibull distribution.³⁰ In addition, the neural network and the genetic algorithm were used to get the best smoothing parameter of the three-parameter Weibull distribution.³¹

By comparing the performance of 12 methods, Teimouri *et al.* illustrated that the parameters estimated by the different methods differ from each other considerably even in large-sized sample situations.³² Maurizio and Penta argued that the technical usefulness of the classical (frequentist) statistical analysis is very limited when the sample size is small.³³ Hence, developing accurate, robust, and efficient parameter estimation methods are still of great significance.

2. Minimum Discrepancy Principle for Scale Parameter Estimation

2.1. A pseudo estimator for the scale parameter of the Weibull distribution

The probability density function and cumulative distribution function of a three-parameter Weibull distributed random variable T are

$$f(t) = \frac{\beta}{\eta} \left(\frac{t - \gamma}{\eta} \right)^{\beta-1} \exp \left[- \left(\frac{t - \gamma}{\eta} \right)^{\beta} \right], \quad (2.1)$$

$$F(t) = 1 - \exp \left[- \left(\frac{t - \gamma}{\eta} \right)^\beta \right], \quad (2.2)$$

where β , η and γ stand for the shape parameter, scale parameter and location parameter, respectively.

By Eq. (2.2), a mapping from t and $F(t)$ to η can be constructed:

$$\eta = (t - \gamma) / (-\ln(1 - F(t)))^{1/\beta}. \quad (2.3)$$

By Eq. (2.3), the Weibull scale parameter can be estimated with a value of the random variable t and the corresponding cumulative distribution probability $F(t)$, given the location parameter γ and the shape parameter β .

When a sample of size n is available, let $t_{1,n}, t_{2,n}, \dots, t_{n,n}$ denote the sample values arranged in ascending order (i.e. $t_{1,n} \leq t_{2,n} \leq \dots \leq t_{n,n}$), the corresponding cumulative distribution probabilities can be estimated by the median rank estimator:

$$F(t_{i,n}) = \frac{i - 0.3}{n + 0.4}. \quad (2.4)$$

Therefore, through Eqs. (2.3) and (2.4), the scale parameter can be estimated with a sample value $t_{i,n}$, provided that the location parameter γ and shape parameter β are known. In this sense, $(T_{i,n} - \gamma) / (-\ln(1 - \frac{i-0.3}{n+0.4}))^{1/\beta}$ can be regarded as a pseudo estimator for the scale parameter, i.e. a scale parameter value can be estimated with a sample value $t_{i,n}$ ($i = 1, 2, \dots, n$) as

$$\eta_i = \frac{t_{i,n} - \gamma}{(-\ln(1 - F(t_{i,n})))^{1/\beta}}. \quad (2.5)$$

2.2. Scale parameter estimate variation

Since the sample value $t_{i,n}$ is not exactly the distribution percentile corresponding to the cumulative distribution probability estimated by the median rank estimator, the scale parameters estimated by Eq. (2.5) with different sample values are different from each other. Besides, the differences between the scale parameter values estimated by Eq. (2.5) also depend on the error of the location/shape parameter applied, as the true parameter values are unknown. It is easy to know that the larger the error of the location/shape parameter, the larger the differences between the estimated scale parameter values η_i ($i = 1, 2, \dots, n$).

2.2.1. The effect of location parameter error

Let $\Delta\gamma$ denote the error of the location parameter applied in Eq. (2.5), the variation of the estimated scale parameter resulted from $\Delta\gamma$ is

$$\Delta\eta_i(\Delta\gamma) = \eta_i(\gamma + \Delta\gamma, \beta) - \eta_i(\gamma, \beta) = \Delta\gamma / (-\ln(1 - F(t_{i,n})))^{1/\beta}, \quad (2.6)$$

where $\eta_i(\gamma + \Delta\gamma, \beta)$ stands for the scale parameter estimated by Eq. (2.5) with sample value $t_{i,n}$ as well as location parameter value $\gamma + \Delta\gamma$ and shape parameter

value β , $\eta_i(\gamma, \beta)$ stands for that with sample value $t_{i,n}$ as well as location parameter value γ and shape parameter value β .

Because the $F(t_{i,n})$ applied in Eq. (2.6) is a function of the ordinal number i (ref. Eq. (2.4)), the individual $\Delta\eta_i(\Delta\gamma)$ ($i = 1, 2, \dots, n$) are different from each other. The maximal difference between $\Delta\eta_i(\Delta\gamma)$ ($i = 1, 2, \dots, n$), which is also the maximal difference between the scale parameter estimates η_i ($i = 1, 2, \dots, n$), estimated with the individual sample values $t_{i,n}$ ($i = 1, 2, \dots, n$) is directly proportional to the location parameter error $\Delta\gamma$, i.e.

$$\begin{aligned} &\Delta\eta_1(\gamma + \Delta\gamma, \beta) - \Delta\eta_n(\gamma + \Delta\gamma, \beta) \\ &= \Delta\eta_1(\Delta\gamma) - \Delta\eta_n(\Delta\gamma) \\ &= \frac{\Delta\gamma}{(-\ln(1 - F(t_{1,n})))^{\frac{1}{\beta}}} - \frac{\Delta\gamma}{(-\ln(1 - F(t_{n,n})))^{\frac{1}{\beta}}}. \end{aligned} \tag{2.7}$$

Shown in Fig. 1 is the situation of 15 sample values from W (2.0, 1 000, 1 000) (Weibull distribution with shape parameter 2.0, scale parameter 1 000 and location parameter 1 000), where, the ordinate is $(\eta_1(\gamma + \Delta\gamma, \beta) - \eta_{15}(\gamma + \Delta\gamma, \beta))^2$, i.e. the squared maximum difference between the estimated scale parameters resulted from location parameter error $\Delta\gamma$.

2.2.2. The effect of shape parameter error

Let $\Delta\beta$ denote the error of the shape parameter applied in Eq. (2.5), the corresponding variation of the estimated scale parameter η_i equals

$$\Delta\eta_i(\Delta\beta) = \frac{t_{i,n} - \gamma}{(-\ln(1 - F(t_{i,n})))^{\frac{1}{\beta + \Delta\beta}}} - \frac{t_{i,n} - \gamma}{(-\ln(1 - F(t_{i,n})))^{\frac{1}{\beta}}}. \tag{2.8}$$

These variations come from both the error of the shape parameter and the uncertainty of the sample values (different sample values correspond to different errors in the corresponding cumulative distribution probabilities estimated by the median rank estimator). The difference between the estimated $\eta_1(\gamma, \beta + \Delta\beta)$ and

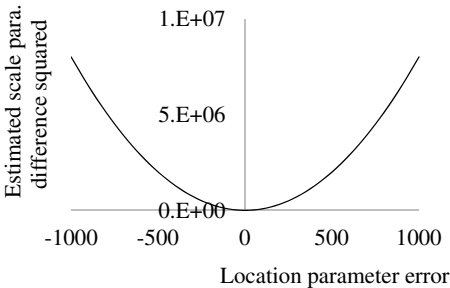


Fig. 1. Squared difference between estimated scale parameters with respect to the different location parameter error $\Delta\gamma$.

$\eta_{15}(\gamma, \beta + \Delta\beta)$, i.e. the maximum difference between the estimated scale parameters, resulted from shape parameter error $\Delta\beta$ equals

$$\begin{aligned} & \eta_1(\gamma, \beta + \Delta\beta) - \eta_{15}(\gamma, \beta + \Delta\beta) \\ &= \frac{t_{p1} - \gamma}{(-\ln(1 - F(t_{p1})))^{\frac{1}{\beta + \Delta\beta}}} - \frac{t_{p15} - \gamma}{(-\ln(1 - F(t_{p15})))^{\frac{1}{\beta + \Delta\beta}}}, \end{aligned} \quad (2.9)$$

where t_{p1} and t_{p15} stand for the exact $p_1 \times 100$ th ($p_1 = (1 - 0.3)/(15 + 0.4)$) and $p_{15} \times 100$ th ($p_{15} = (15 - 0.3)/(15 + 0.4)$) percentiles of the Weibull distribution, respectively.

The squared difference $\eta_1(\gamma, \beta + \Delta\beta) - \eta_{15}(\gamma, \beta + \Delta\beta)$ resulted from the shape parameter error $\Delta\beta$ in the situation of a sample of size 15 from $W(2.0, 1\,000, 1\,000)$ is shown in Fig. 2. It demonstrates that the larger the shape parameter error, the larger the difference between the scale parameter values estimated with the different sample values.

2.3. Principle and procedure to estimate Weibull distribution parameters

As the larger error in the location/shape parameter results in larger difference between the scale parameter estimates associated with the individual sample values, the right location parameter and shape parameter can be identified by a two-dimensional recursion process, with regards to location parameter and shape parameter, respectively. The standard deviation of the n scale parameter estimates $\eta_1, \eta_2, \dots, \eta_n$ obtained from the n sample values $t_{1,n}, t_{2,n}, \dots, t_{1,n}$, denoted by σ_η , is used to characterize the difference. For parameter estimation, an initial shape parameter value β_0 (a possible minimum shape parameter value) and a suitable increment $\Delta\beta$ are set first, and a series of shape parameter values $\beta_j = \beta_0 + j\Delta\beta$ ($j = 0, 1, 2, \dots$) are considered. To each of the shape parameter value β_j , a series of location parameter values $\gamma_k = k\Delta\gamma$ ($\Delta\gamma$ is an increment of location parameter γ used in the recursion process, $k = 0, 1, 2, \dots$) are tried. Obviously, a pair of (β_j, γ_k)

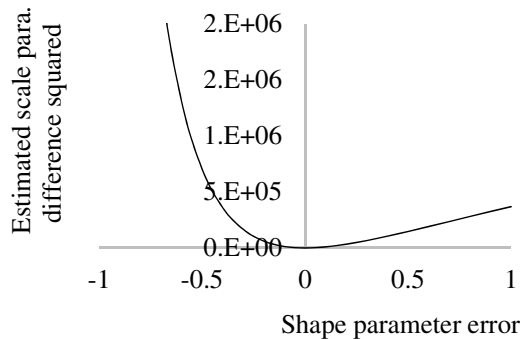


Fig. 2. Squared difference between estimated scale parameters $\eta_1(\gamma, \beta + \Delta\beta)$ and $\eta_{15}(\gamma, \beta + \Delta\beta)$ with respective to the different shape parameter error $\Delta\beta$.

far from the true parameter values will yield a large σ_η . The pair of (β_j, γ_k) obtained from minimizing the standard deviation σ_η are the estimated shape parameter $\hat{\beta}$ and location parameter $\hat{\gamma}$.

When the right shape parameter and location parameter are obtained, a scale parameter can be estimated by means of Eq. (2.5) with a sample value. As mentioned above, since the sample value $t_{i,n}$ is not exactly the $(i - 0.3)/(n + 0.4) \times 100$ th percentile of the distribution, the scale parameter values estimated by Eq. (2.5) with the individual sample values will differ from each other. In other words, the different pairs of $(t_{i,n}, F(t_{i,n}))$ yield different scale parameter estimates even though the right shape parameter and location parameter are used. To reduce the effect of sample uncertainty, the average of such n estimates $\eta_i (i = 1, 2, \dots, n)$ is taken as the final estimate of the scale parameter:

$$\eta = \frac{1}{n} \sum_{i=1}^n \frac{t_{i,n} - \hat{\gamma}}{(-\ln(1 - F(t_{i,n})))^{1/\hat{\beta}}}. \quad (2.10)$$

3. Weibull Distribution Parameter Estimation

3.1. Procedures for Weibull distribution parameter estimation

The procedures to estimate the three parameters of the Weibull distribution according to a sample of size 15 are described below. The 15 sample values, 1 372.9, 1 464.2, 1 980.6, 1 899.9, 1 716.2, 2 097.6, 2 105.0, 1 615.6, 1 657.6, 1 687.8, 2 170.8, 2 446.1, 2 274.7, 1 573.9 and 2 639.3, are generated by Monte Carlo sampling from the Weibull distribution $W(2.0, 1\ 000, 1\ 000)$. To roughly estimate the parameters, five potential shape parameter values, 1.0, 1.5, 2.0, 3.0, and 4.0, are considered (smaller initial value and smaller increment are necessary for higher accuracy). To each of the shape parameter value, a series of potential location parameter values, 0, 50, 100, ... (smaller increment should be used for higher accuracy), are tried to get a series of standard deviations of the scale parameter estimates $\eta_i(\beta_j, \gamma_k) (i = 1, 2, \dots, 15)$ estimated by means of Eq. (2.5) with the 15 sample values, respectively. For each shape parameter value β_j and a series of location parameter values $\gamma_k (k = 1, 2, \dots)$, a series of $\sigma_\eta(\beta_j, \gamma_k) (k = 1, 2, \dots)$ can be obtained. The results are shown in Fig. 3, where each curve represents a relationship between $\sigma_\eta(\beta_j, \gamma_k) (k = 1, 2, \dots, 15)$ and γ_k for a particular shape parameter value β_j . A conditional minimal standard deviation of the estimated scale parameters $\sigma_{\eta, \min}(\beta_j)$ can be obtained for each of the β_j . The minimal standard deviation of the estimated scale parameters, i.e. $\text{Min}\{\sigma_{\eta, \min}(\beta_j), j = 1, 2, \dots, J\}$ (J is the total number of the shape parameter values considered), denoted by $\sigma_{\eta, \min}$, is achieved at $\beta = 2.0, \gamma = 1\ 200$. Therefore, the shape parameter is estimated as 2.0 and the location parameter is estimated as 1 200. Based on the estimated shape parameter and location parameter, the scale parameter is estimated by Eq. (2.10) as 806.0.

Comparing with the true values of the Weibull distribution parameters, i.e. $\beta = 2.0, \eta = 1\ 000$ and $\gamma = 1\ 000$, the relative errors are 0%, 20% and -19.4% for the estimated shape parameter, location parameter and scale parameter, respectively.

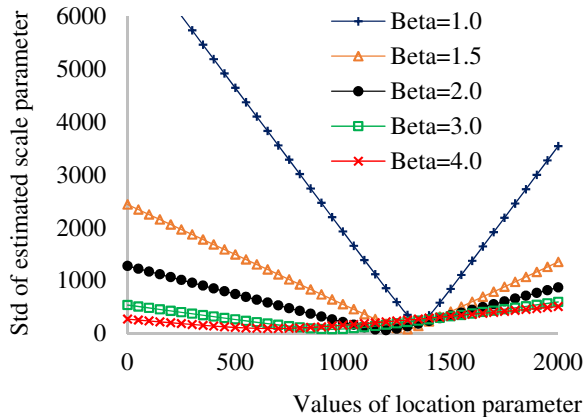


Fig. 3. Standard deviations of the estimated scale parameter values with respect to the different shape parameter Beta and location parameter Gamma values.

3.2. Minimum discrepancy criterion

As described above, n scale parameter values can be obtained, respectively, with n sample values by Eq. (2.5) with given shape parameter and location parameter. Besides the fact that the uncertainty of the sample values makes the scale parameter estimates differ from each other, the errors of the shape parameter and location parameter used in Eq. (2.5) bring about additional discrepancy between such estimated scale parameter values. For a given parameter value β , a minimal standard deviation of the estimated scale parameter values (denoted by $\sigma_{\eta, \min}(\beta)$) exists at a certain location parameter value. Shown in Fig. 4 is a typical case for a sample of size 15. Figure 4(a) is the relationship between $\sigma_{\eta, \min}(\beta)$ and the shape parameter β . Figure 4(b) is the decrease rate (the slope) of $\sigma_{\eta, \min}(\beta)$ with respect to the shape parameter β . The decrease rate is numerically defined as

$$\frac{\partial \sigma_{\min}(\beta)}{\partial \beta} = \frac{\sigma_{\min}(\beta) - \sigma_{\min}(\beta + \Delta\beta)}{\Delta\beta}. \quad (3.1)$$

As $\sigma_{\eta, \min}(\beta)$ gets its minimum at $\partial \sigma_{\eta, \min}(\beta) / \partial \beta = 0$, the “minimum discrepancy criterion” for scale parameter estimation is expressed as

$$\frac{\partial \sigma_{\min}(\beta)}{\partial \beta} = 0. \quad (3.2)$$

Based on this criterion, the Weibull distribution parameters β and γ minimizing the standard deviation of the estimated scale parameter values $\sigma_{\eta}(\beta, \gamma)$ can be found by a recursion process (e.g. using a shape parameter increment $\Delta\beta = 0.1$, starting at an initial value 0.1, and a location parameter increment $\Delta\gamma = 50$, starting at 0). The dotted line in Fig. 4(c) is the fitted relationship between $\partial \sigma_{\eta, \min}(\beta) / \partial \beta$ and β , where the built-in cubic polynomial function in EXCEL, Microsoft Office, is used. To obtain a high correlation coefficient for the fitted curve, part of the $\partial \sigma_{\eta, \min}(\beta) / \partial \beta$ data

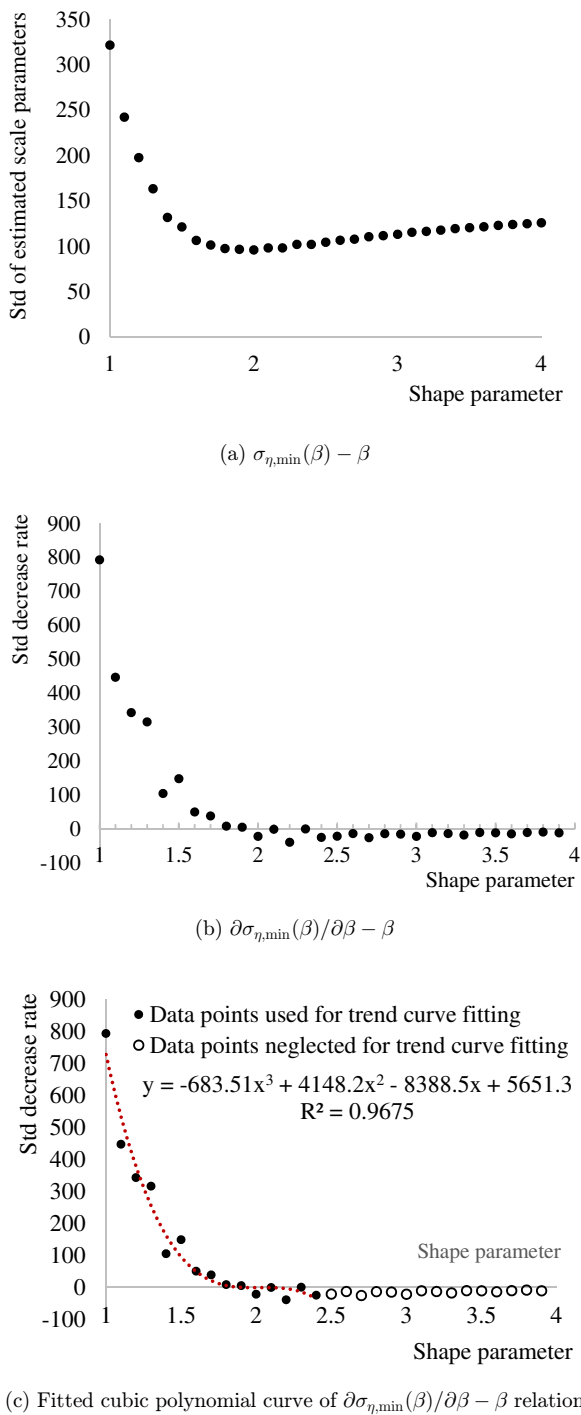


Fig. 4. Relationships produced during parameter estimation process.

points corresponding to larger shape parameter values are removed. For the situation demonstrated in Fig. 4, the Weibull shape parameter is estimated as $\beta = 1.896$. Correspondingly, the location parameter is obtained by recalling the recursion process as $\gamma = 1000$, and the scale parameter is obtained by Eq. (2.10) as $\eta = 980.7$.

The parameter estimation method described above is referred to the minimum discrepancy method (MD method) in this paper, and the pertinent estimate is abbreviated as MDE below. In the parameter estimation method, the zero-decrease rate criterion, rather than a minimal standard deviation criterion, is applied, since a great number of case studies demonstrate that the zero-decrease rate criterion is more robust against sample uncertainty.

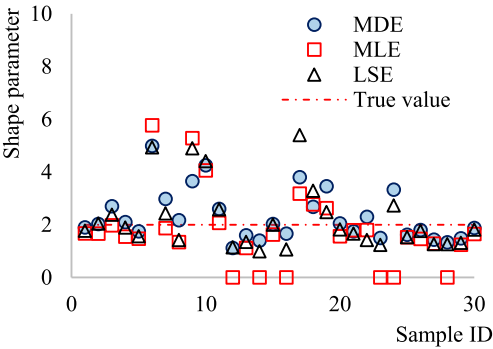
3.3. Parameter estimation results according to samples of size 15

To compare the MD method with the conventional methods (maximum likelihood method and Weibull plot correlation coefficient-based least squares method, denoted by ML method and LS method, respectively), 45 samples are used. Shown in Fig. 5 are the parameter estimation results of 30 samples of size 15, generated by Monte Carol sampling from $W(2.0, 1000, 1000)$, where MDE, MLE and LSE stand for the estimates of the MD method, ML method and LS method, respectively. Among the 30 samples, there are six cases where ML method fails to get solution. These cases are marked with zero parameter values in Fig. 5.

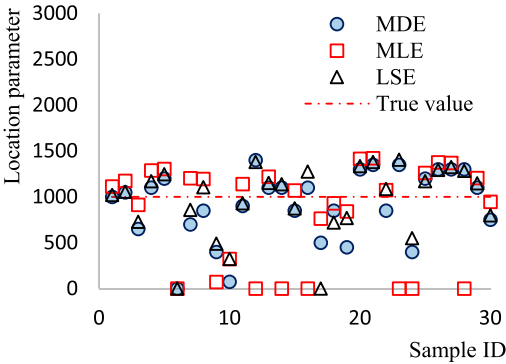
The parameter estimation results show that, in more than two-thirds of the samples, the shape parameters estimated by MD method are closer to the true value; most of the location parameters estimated by the three methods do not differ from each other considerably, while most of the location parameters estimated by MD method are smaller than those estimated by ML method and LS method (in other words, MD method is more conservative); most of the scale parameters estimated by the three methods are close to each other, while almost all the scale parameters estimated by MD method are greater than those estimated by the other two methods (i.e. MD method is more conservative in sense of the degree of dispersion of the estimated Weibull distribution).

Besides, the robustness of MD method is better than the other two methods. Shown in Table 1 are the ranges, the mathematical expectations (the means) and the standard deviations (Std) of the parameters estimated by the different methods for the 30 samples.

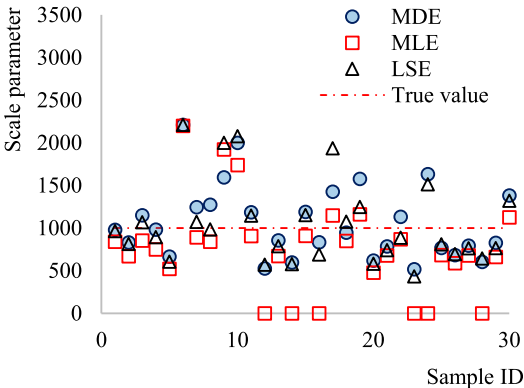
Since the ultimate aim of parameter estimation is to acquire the probability density function (pdf) or cumulative distribution function (cdf), it is more straightforward and more meaningful to evaluate the parameter estimate results in terms of the pdf or cdf characterized by the parameter estimates, instead of the individual parameter estimates themselves. As for the application of a pdf or cdf in product reliability or failure probability calculation, the most significant is the left tail of the pdf curve or the low probability part of the cdf curve. Shown in Fig. 6 are the low probability parts (from 0 to 0.1) of the cdf curves estimated by the three



(a) Shape parameters estimated by different methods



(b) Location parameters estimated by different methods

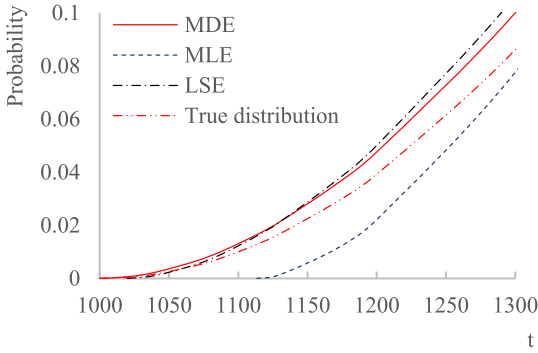


(c) Scale parameters estimated by different methods

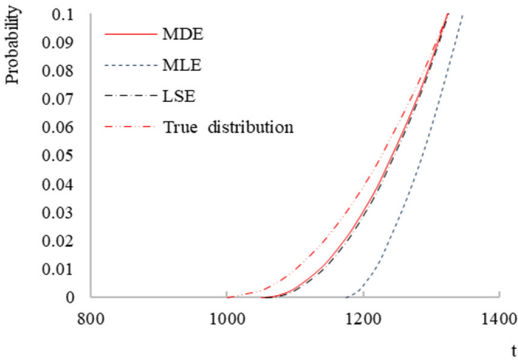
Fig. 5. Weibull distribution parameters estimated by different methods ($n = 15$).

Table 1. Statistical property of the estimated parameters.

Para.	Measure	MDE	MLE	LSE
β	Range	1.12–4.99	1.12–5.77	0.99–5.4
	Mean	2.31	2.18	2.19
	Std	0.95	1.23	1.22
η	Range	516.5–2 209.5	478.2–2 199	432.3–2 210.8
	Mean	1 060.5	943.3	1 034.0
	Std	431.6	434.1	477.4
γ	Range	0–1 400	0–1 422.1	0–1 405.1
	Mean	915.8	1 025.9	966.8
	Std	380.0	390.8	384.7

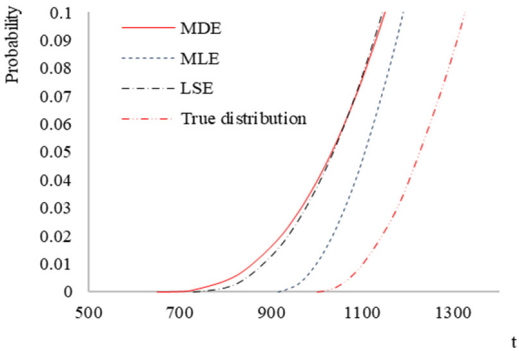


(a)

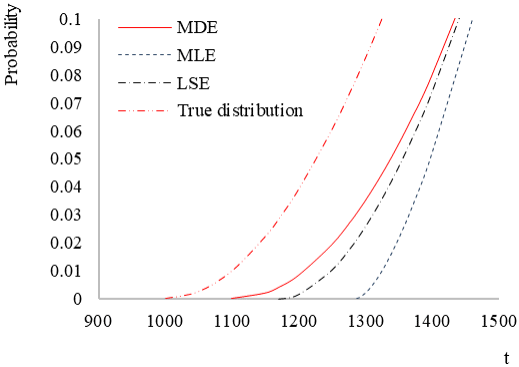


(b)

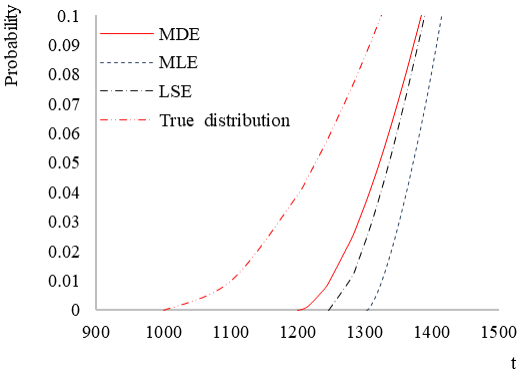
Fig. 6. True Weibull distribution curve (low probability part) and those estimated by different methods. (a) LSE is the best in precision, while it is non-conservative in the very low probability part, MLE is the worst, non-conservative, (b) MDE is a little bit better than LSE, much better than MLE, (c) MLE is the best, LSE is a little bit better than MDE, (d) MDE is the best, LSE is medium, MLE is the worst, all methods are non-conservative, (e) MDE is the best, LSE is medium, MLE is the worst, all are non-conservative and (f) Eight probability distribution curves (parts) estimated, respectively, by the three methods.



(c)



(d)



(e)

Fig. 6. (Continued)

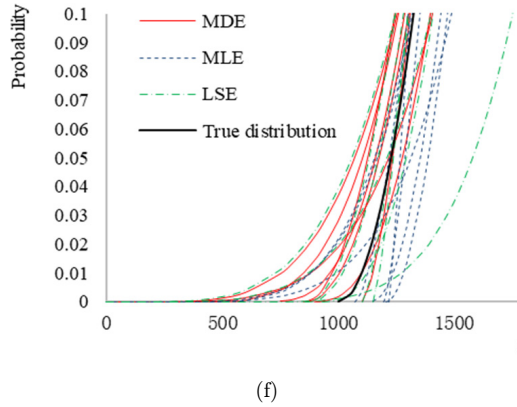


Fig. 6. (Continued)

methods, respectively, with the first 15 samples in the total 30, except for the two samples where ML method fails to obtain parameter estimate.

For the first sample (Fig. 6(a)), LSE is the best in precision in the very low probability part, but not conservative in estimating the failure probability; MDE is not so close to the true curve in the very low probability part as LSE, but it is conservative for failure probability estimation; MLE is the worst and non-conservative. For the second sample (Fig. 6(b)), MDE and LSE are very close to each other, MLE is far from them, all the three methods are not conservative. For the third sample (Fig. 6(c)), MLE is the best, MDE is the worst, all the three methods are conservative. For the fourth (Fig. 6(d)) and fifth samples (Fig. 6(e)), MDE is the best, LSE is medium, MLE is the worst, all the three methods are non-conservative. Other eight cases (samples 6–15) are shown in Fig. 6(f), demonstrating that MDE outperforms MLE and LSE overall.

A summary is listed in Table 2. In the 30 samples of size 15, there are 11 samples to which the MD method achieves the best results, 14 samples to which the MD method achieves medium results, and 5 samples to which the MD method achieves the worst results; there are 10 samples to which LSE achieves the best results, 15 samples to which LSE achieves medium results, and 5 samples to which LSE achieves the worst results; there are 9 samples to which MLE achieves the best results, 1 sample to which MLE achieves medium result, and 20 samples to which MLE achieves the worst results (including 6 samples to which no solution can be obtained). Among the 30 samples, there are 15 samples to which the MD method achieves conservative results, 12 samples to which LS method achieves conservative results, and 8 samples to which ML method achieves conservative results.

Comprehensively evaluated in terms of accuracy, conservatism, and robustness of the estimation results of the 30 samples, the MD method outperforms the two conventional methods. Besides, the MD method is much easier to apply.

Table 2. Summary of different parameter estimation methods.

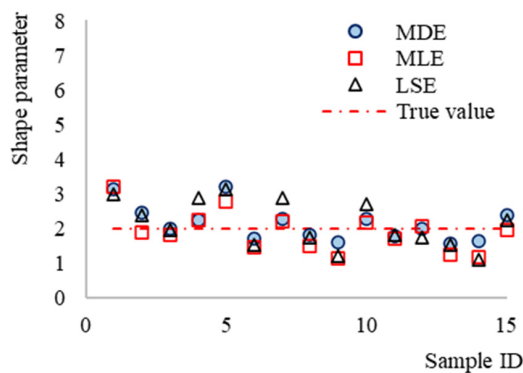
Sample ID	MDE	LSE	MLE
1	M/C	B/N ⁺	W/N
2	B/N	M/N	W/N
3	W/C	M/C	B/C
4	B/N	M/N	W/N
5	B/N	M/N	W/N
6	M/C ⁻	W/C ⁻	B/C ⁻
7	M/C	B/C	W/N
8	M/C	B/N ⁺	W/N
9	M/C	W/C	B/C ⁻
10	M/C	W/C	B/C
11	B/C ⁻	M/N	W/N
12	M/N	B/N	—
13	B/N ⁺	M/N ⁺	W/N
14	B/N ⁺	M/N ⁺	—
15	W/C	M/C	B/N ⁺
16	B/N	M/N	—
17	M/C	W/C	B/C
18	M/C	W/C	B/C
19	W/C	M/C	B/C ⁻
20	B/N	M/N	W/N
21	B/N	M/N	W/N
22	W/C	B/N ⁺	M/N
23	M/N	B/N ⁺	—
24	M/C	B/C	—
25	M/N	B/N	W/N
26	M/N	B/N	W/N
27	B/N	M/N	W/N
28	M/N	B/N	—
29	B/N ⁺	M/N ⁺	W/N ⁺
30	W/C	M/C	B/C

Notes: B-best, M-medium, W-worst, C-conservative, N-non-conservative, C⁻-conservative in very low probability part but non-conservative in higher probability part, N⁺-non-conservative in very low probability part but conservative in higher probability part.

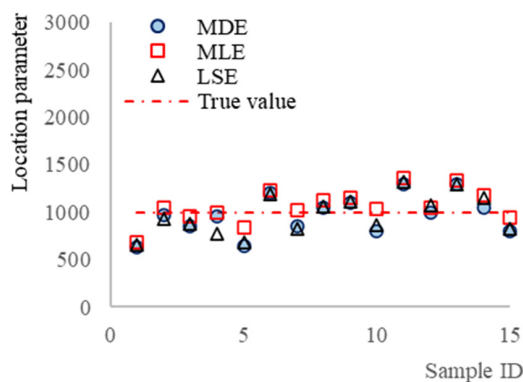
4. Method Comparison in More Situations

4.1. Weibull distribution parameter estimation results of samples of size 30

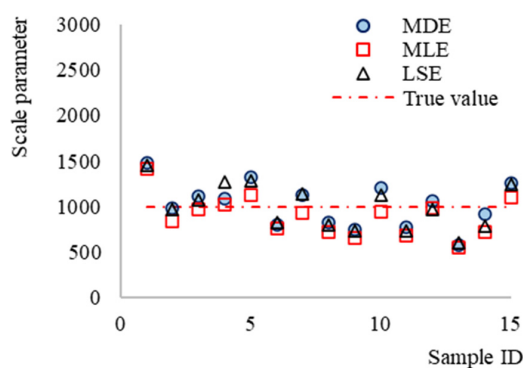
Shown in Fig. 7 are the parameter estimates of 15 samples of size 30, generated by Monte Carol sampling from $W(2.0, 1000, 1000)$. For the larger sample situations, the three parameters estimated by the three methods (i.e. MD method, ML method and LS method) are very close to each other. The robustness of MD method is slightly better than the two conventional methods, though the differences are not very significant. Shown in Table 3 are the ranges, the mathematical expectations and the standard deviations of the Weibull parameters estimated by the different methods with the 15 samples of size 30.



(a) Weibull shape parameters estimated by different methods



(b) Weibull location parameters estimated by different methods



(c) Weibull scale parameters estimated by different methods

Fig. 7. Weibull distribution parameters estimated by different methods ($n = 30$).

Table 3. Statistical property of the estimated parameters ($n = 30$).

Para.	Measure	MDE	MLE	LSE
β	Range	1.58–3.20	1.15–3.20	1.11–3.13
	Mean	2.15	1.93	2.13
	Std	0.51	0.58	0.67
η	Range	580.6–1 479.1	553.7–1 414.8	596.9–1 450.7
	Mean	966.7	1 063.32	975.9
	Std	208.7	176.3	211.6
γ	Range	625–1 300	681.5–1 356	655.1–1 315.7
	Mean	1 017.4	894.5	999.7
	Std	245.6	223.5	250.1

4.2. Weibull distribution parameter estimation results of samples from $W(3.0, 1\,000, 1\,000)$

Shown in Fig. 8 are the parameter estimates of 10 samples of size 15, randomly generated from $W(3.0, 1\,000, 1\,000)$. The results demonstrate that MDE method

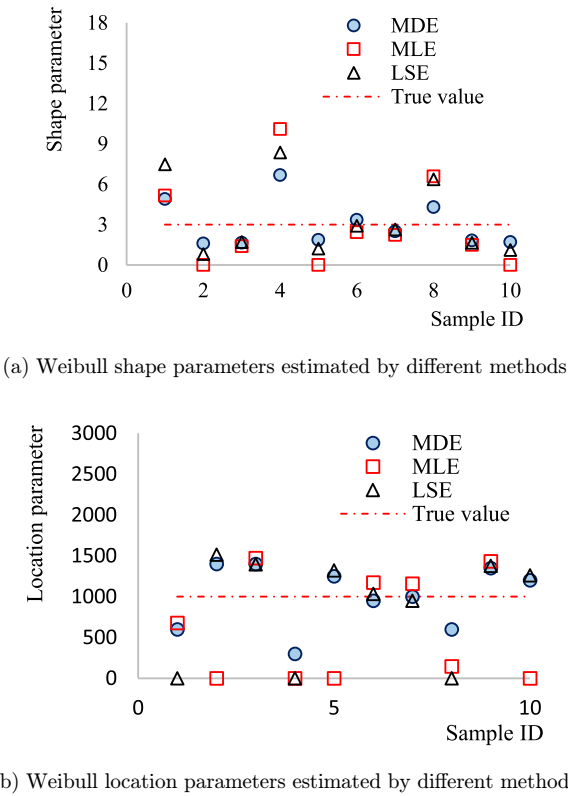
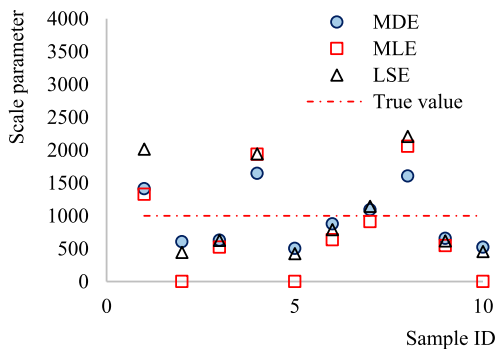


Fig. 8. Weibull distribution parameters estimated by different methods ($W(3.0, 1\,000, 1\,000)$, sample size $n = 15$).

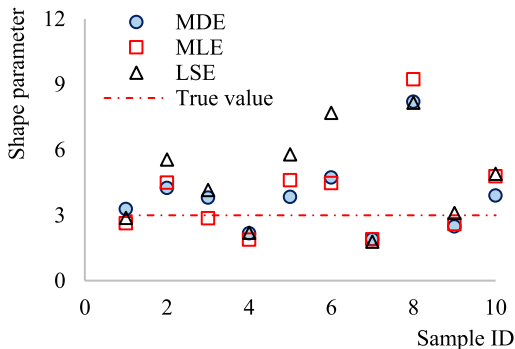


(c) Weibull scale parameters estimated by different methods

Fig. 8. (Continued)

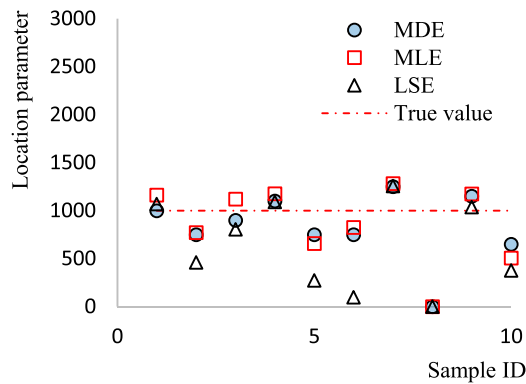
Table 4. Statistical property of the estimated parameters.

Para.	Measure	MDE	MLE	LSE
β	Range	1.60–6.68	1.39–10.1	0.81–8.36
	Mean	3.04	2.94	3.43
	Std	1.74	3.34	2.86
η	Range	503.5–1 646.0	521.6–2 057.5	425.6–2 208.3
	Mean	1 142.5	793.25	1 067.0
	Std	503.5	763.746	716.5
γ	Range	300–1 400	0–1 472.4	0–1 514.5
	Mean	1 005	606.2	885.46
	Std	388.3	645.3	633.129

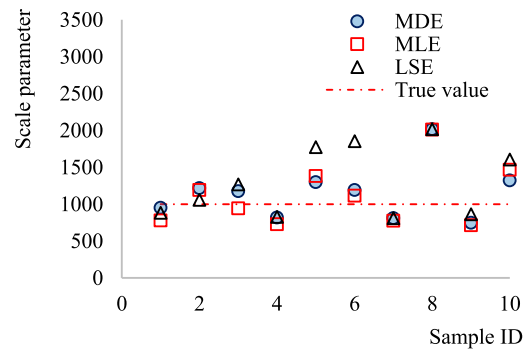


(a) Weibull shape parameters estimated by different methods

Fig. 9. Weibull distribution parameters estimated by different methods ($W(3.0, 1\,000, 1\,000)$, sample size $n = 30$).



(b) Weibull location parameters estimated by different methods



(c) Weibull scale parameters estimated by different methods

Fig. 9. (Continued)

outperforms the two conventional methods. Shown in Table 4 are the ranges, the mathematical expectations and the standard deviations of the Weibull parameters estimated by the different methods with these samples.

Table 5. Statistical property of the estimated parameters.

Para.	Measure	MDE	MLE	LSE
β	Range	1.90–8.20	1.88–9.23	1.79–8.16
	Mean	3.86	3.94	4.62
	Std	1.79	2.18	2.2
η	Range	747.7–2 016.9	713.9–2 011.2	810.1–2 016.3
	Mean	1 154.8	1 110.1	1 296.2
	Std	370.7	417.8	474.2
γ	Range	0–1 250	0–1 281.7	0–1 256
	Mean	830.0	868.3	646.9
	Std	353.7	401.6	458.5

Shown in Fig. 9 are the parameter estimates of 10 samples of size 30, randomly generated from $W(3.0, 1000, 1000)$. The results demonstrate that MD method is superior to the two conventional methods for most of the samples. Shown in Table 5 are the ranges, the mathematical expectations and the standard deviations of the Weibull parameters estimated by the different methods with the 10 samples.

5. Conclusions

A new parameter estimation method, i.e. the minimum discrepancy method (MD method), is proposed for the three-parameter Weibull distribution. This method is constituted with a minimum discrepancy criterion and a recursion algorithm for parameter searching. By means of a pseudo estimator, n different scale parameter values are obtained with the individual sample values in a sample of size n , with given shape parameter and location parameter. The standard deviation of such obtained scale parameter values depends on the shape/location parameter values used in the pseudo estimator. The right shape parameter and location parameter are those minimizing the standard deviation of such estimated scale parameter values. The scale parameter is eventually estimated as the average of the estimated scale parameter values with the individual sample values, based on the right shape parameter and location parameter.

Sixty-five randomly generated samples are used to verify the new parameter estimation method. For the 45 samples from $W(2.0, 1000, 1000)$, MDE (the estimate of the new method) outperforms MLE (the maximum likelihood estimate) and LSE (the least squares estimate). For the 20 samples from $W(3.0, 1000, 1000)$, MDE is much better than MLE and LSE. Overall, the new method has advantages in robustness, conservatism, accuracy, and simplicity in the situations of relatively small size of samples ($n = 15\text{--}30$).

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